

Query Migration Evaluation Results

Correctness (equivalence rating vs. `raw_query`) and relative performance metrics per query, for test-case DBs that have completed `evaluate_queries.py`.

Equivalence scale (weakest to strongest): `INV` < `NEQ` < `DB_EQ` < `EQ`.

Relative metrics are only reported when equivalence is above `NEQ` (i.e. not `INV` / `NEQ`); otherwise the runtime/cost figures aren't meaningful comparisons and are omitted.

Method definitions

Let D' be the real source database and D the LLM-generated target database, where each table $T \in D$ is defined by a mapping $f_T : D' \rightarrow T$ (a `SELECT` over D' , i.e. `table_generation_queries`), so $D = \{f_T(D')\}_{T \in D}$. Let $q \in Q$ be a query written against D .

- raw_query** (ground truth): $q_{raw} = q[T \mapsto f_T]_{T \in D}$ — each table reference in q is substituted, deterministically and exactly, by its defining CTE f_T , producing a query that runs on D' alone with a provably identical result to q on D . This is the reference every other method is checked against.
- baseline_1**: $q'_1 = \text{LLM}(q, \text{schema}(D), \text{schema}(D'), \text{samples}(D'))$ — the LLM is given q and the schemas of D and D' (plus sample rows of D') but **not** f , and must infer the correspondence between D and D' itself before producing a query on D' .
- baseline_2**: $q'_2 = \text{LLM}(q, \text{schema}(D), \text{schema}(D'), \{f_T\}_{T \in D})$ — same as `baseline_1`, except the LLM is additionally handed the true mapping $\{f_T\}$, so it only has to apply a known correspondence rather than discover one.
- optim_query**: $q_{opt} = \text{LLM}(q_{raw})$ — not a migration at all; the LLM is asked to produce an optimized/rewritten form of `raw_query` itself (already a D' -only query), so it measures rewrite quality rather than cross-database correspondence.

db3532

Correctness (Equivalence)

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	EQ	NEQ	DB_EQ	DB_EQ
2	ERROR: Failed to measure raw_query: canceling statement due to stat	-	-	-
3	EQ	NEQ	DB_EQ	DB_EQ

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
4	EQ	NEQ	DB_EQ	DB_EQ
5	EQ	NEQ	DB_EQ	DB_EQ
7	EQ	DB_EQ	DB_EQ	DB_EQ
8	EQ	NEQ	DB_EQ	DB_EQ
9	EQ	NEQ	DB_EQ	NEQ
10	EQ	-	-	-
11	EQ	NEQ	DB_EQ	DB_EQ
12	EQ	NEQ	DB_EQ	DB_EQ
13	EQ	-	-	-
15	EQ	NEQ	DB_EQ	DB_EQ

Note: Queries 10, 13 are blank for `baseline_1` / `baseline_2` / `optim_query` because their record has `error: "Raw query result does not match source query"`, and `baseline_1_query` / `baseline_2_query` / `optim_query` are `null` — those stages never ran for them.

Relative Cost

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	0.7077	0.2178
2	ERROR: Failed to measure raw_query: canceling statement due to stat	-	-	-
3	1.0000	-	0.9917	0.9801
4	1.0000	-	1.0000	0.5808
5	1.0000	-	0.3671	0.3661
7	1.0000	0.9227	1.7449	1.6611
8	1.0000	-	0.9504	0.8765
9	1.0000	-	0.9635	-
10	1.0000	-	-	-
11	1.0000	-	0.5052	4.2881

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
12	1.0000	-	1.0000	0.5367
13	1.0000	-	-	-
15	1.0000	-	1.0000	1.0000

Relative Cold Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	0.9580	0.9052
2	ERROR: Failed to measure raw_query: canceling statement due to stat	-	-	-
3	1.0000	-	0.9811	1.1649
4	1.0000	-	0.9963	0.2697
5	1.0000	-	0.3341	0.3685
7	1.0000	0.7711	0.8262	0.7695
8	1.0000	-	0.9617	1.2733
9	1.0000	-	0.9425	-
10	1.0000	-	-	-
11	1.0000	-	0.4526	0.4415
12	1.0000	-	1.0184	0.2934
13	1.0000	-	-	-
15	1.0000	-	1.0259	0.9899

Relative Hot Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	0.9598	0.9078
2	ERROR: Failed to measure raw_query: canceling statement due to stat	-	-	-
3	1.0000	-	0.9843	0.5397

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
4	1.0000	-	1.0025	0.2026
5	1.0000	-	0.3059	0.3022
7	1.0000	0.7899	0.9139	0.7818
8	1.0000	-	1.0392	0.6870
9	1.0000	-	0.9629	-
10	1.0000	-	-	-
11	1.0000	-	0.4204	0.4162
12	1.0000	-	0.9991	0.2485
13	1.0000	-	-	-
15	1.0000	-	1.0049	1.0086

db3631

Correctness (Equivalence)

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	EQ	NEQ	NEQ	NEQ
2	EQ	DB_EQ	DB_EQ	DB_EQ
4	EQ	NEQ	INV	INV
5	ERROR: Failed to measure raw_query: connection failed: connection t	-	-	-
6	EQ	DB_EQ	DB_EQ	DB_EQ
7	EQ	DB_EQ	DB_EQ	DB_EQ
9	EQ	DB_EQ	DB_EQ	DB_EQ
10	EQ	NEQ	DB_EQ	DB_EQ
11	EQ	DB_EQ	DB_EQ	DB_EQ
12	EQ	-	-	-
13	EQ	DB_EQ	DB_EQ	DB_EQ
14	EQ	DB_EQ	DB_EQ	DB_EQ

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
15	EQ	DB_EQ	DB_EQ	DB_EQ

Note: Query 12 is blank for baseline_1 / baseline_2 / optim_query because its record has error: "Raw query result does not match source query" , and baseline_1_query / baseline_2_query / optim_query are null — those stages never ran for it.

Relative Cost

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	-	-
2	1.0000	0.0059	0.0064	0.0066
4	1.0000	-	-	-
5	ERROR: Failed to measure raw_query: connection failed: connection t	-	-	-
6	1.0000	0.8838	0.8158	1.1788
7	1.0000	0.0002	0.0003	0.0001
9	1.0000	0.0499	0.0577	0.1085
10	1.0000	-	0.7078	0.7744
11	1.0000	3.9408	0.0005	0.0005
12	1.0000	-	-	-
13	1.0000	0.8221	0.8348	0.8695
14	1.0000	0.6687	0.7388	0.8879
15	1.0000	0.7747	0.7904	0.7982

Relative Cold Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	-	-
2	1.0000	0.1224	0.1631	0.1622
4	1.0000	-	-	-

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
5	ERROR: Failed to measure raw_query: connection failed: connection t	-	-	-
6	1.0000	0.7063	1.1995	1.2142
7	1.0000	0.3323	0.4821	0.0605
9	1.0000	0.1146	0.1761	0.2738
10	1.0000	-	0.7149	0.7903
11	1.0000	0.0023	0.0015	0.0015
12	1.0000	-	-	-
13	1.0000	0.6819	0.8412	0.7223
14	1.0000	0.5277	0.9152	1.0089
15	1.0000	0.9917	1.0585	1.0671

Relative Hot Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	-	-	-
2	1.0000	0.0669	0.1089	0.1035
4	1.0000	-	-	-
5	ERROR: Failed to measure raw_query: connection failed: connection t	-	-	-
6	1.0000	0.8325	1.5351	1.5553
7	1.0000	0.2312	0.3355	0.0613
9	1.0000	0.0872	0.1494	0.2172
10	1.0000	-	0.8488	1.0257
11	1.0000	0.0002	0.0002	0.0002
12	1.0000	-	-	-
13	1.0000	0.8138	1.0061	0.8640
14	1.0000	0.5295	0.8776	0.9674

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
15	1.0000	1.1456	1.1738	1.2753

db3632

Correctness (Equivalence)

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	EQ	DB_EQ	DB_EQ	-
2	EQ	NEQ	DB_EQ	-
3	EQ	NEQ	DB_EQ	-
4	EQ	NEQ	DB_EQ	-
5	EQ	NEQ	DB_EQ	-
7	EQ	INV	DB_EQ	-
8	EQ	NEQ	DB_EQ	-
9	EQ	DB_EQ	DB_EQ	-
10	EQ	NEQ	DB_EQ	-
11	EQ	-	-	-
12	EQ	DB_EQ	DB_EQ	-
13	EQ	DB_EQ	DB_EQ	-
15	EQ	DB_EQ	DB_EQ	-

Note: Query 11 is blank for `baseline_1` / `baseline_2` / `optim_query` because its record has error: "Raw query result does not match source query", and `baseline_1_query` / `baseline_2_query` / `optim_query` are `null` — those stages never ran for it.

Relative Cost

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	1.0812	0.9785	-
2	1.0000	-	1.0213	-
3	1.0000	-	0.9655	-
4	1.0000	-	0.9791	-
5	1.0000	-	0.9973	-

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
7	1.0000	-	0.9774	-
8	1.0000	-	0.0268	-
9	1.0000	0.9988	0.9988	-
10	1.0000	-	0.9841	-
11	1.0000	-	-	-
12	1.0000	12.1246	0.9924	-
13	1.0000	0.4057	0.7149	-
15	1.0000	3.1143	1.0249	-

Relative Cold Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	1.7216	0.9849	-
2	1.0000	-	1.0700	-
3	1.0000	-	0.7363	-
4	1.0000	-	1.0292	-
5	1.0000	-	1.1953	-
7	1.0000	-	1.1416	-
8	1.0000	-	0.0456	-
9	1.0000	0.9509	1.0292	-
10	1.0000	-	0.9785	-
11	1.0000	-	-	-
12	1.0000	1.2370	0.9479	-
13	1.0000	0.0927	0.5217	-
15	1.0000	3.8513	1.0117	-

Relative Hot Runtime

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
1	1.0000	1.3898	0.8475	-
2	1.0000	-	0.9979	-

Query ID	raw_query (ground truth)	baseline_1	baseline_2	optim_query
3	1.0000	-	1.1521	-
4	1.0000	-	1.0257	-
5	1.0000	-	0.9813	-
7	1.0000	-	0.9959	-
8	1.0000	-	0.0399	-
9	1.0000	0.9456	1.0151	-
10	1.0000	-	1.1488	-
11	1.0000	-	-	-
12	1.0000	1.2345	0.9857	-
13	1.0000	0.0620	0.4909	-
15	1.0000	1.4126	0.9838	-

Summary

Per (db, method): **equivalence %** = share of attempted queries rated EQ / DB_EQ against raw_query (N = number of queries where that method has an analysis entry at all); **geomean relative hot runtime** and **geomean relative cost** are the geometric mean of relative_hot_runtime / relative_cost taken only over that method's queries with equivalence above NEQ (i.e. excluding INV / NEQ , since those figures aren't meaningful there) — so each of the three numbers in a cell can be an average over a different N; that N is shown alongside each figure.

DB	raw_query (ground truth)	baseline_1	baseline_2	optim_query
db3532	Eq: 100.0% (N=12) GM hot runtime: 1.0000 (N=12) GM cost: 1.0000 (N=12)	Eq: 10.0% (N=10) GM hot runtime: 0.7899 (N=1) GM cost: 0.9227 (N=1)	Eq: 100.0% (N=10) GM hot runtime: 0.8033 (N=10) GM cost: 0.8547 (N=10)	Eq: 90.0% (N=10) GM hot runtime: 0.4917 (N=9) GM cost: 0.8112 (N=9)
db3631	Eq: 100.0% (N=12) GM hot runtime: 1.0000 (N=12) GM cost: 1.0000 (N=12)	Eq: 72.7% (N=11) GM hot runtime: 0.1309 (N=8) GM cost: 0.1335 (N=8)	Eq: 81.8% (N=11) GM hot runtime: 0.2253 (N=9) GM cost: 0.0624 (N=9)	Eq: 81.8% (N=11) GM hot runtime: 0.1944 (N=9) GM cost: 0.0674 (N=9)

DB	raw_query (ground truth)	baseline_1	baseline_2	optim_query
db3632	Eq: 100.0% (N=13) GM hot runtime: 1.0000 (N=13) GM cost: 1.0000 (N=13)	Eq: 41.7% (N=12) GM hot runtime: 0.6768 (N=5) GM cost: 1.7528 (N=5)	Eq: 100.0% (N=12) GM hot runtime: 0.7265 (N=12) GM cost: 0.7143 (N=12)	-